False Consensus: Why Intermediate Agreement Is Not a Safe Early-Exit Signal for Reasoning LLMs

Anonymous ACL submission

Abstract

A popular way to cut reasoning-model inference cost is to probe the partial trajectory for its current answer and stop once probes *agree*. We show this bet is unsafe, and that the failure is in the *signal*, not the threshold. On frozen trajectories from two models and three math benchmarks, a *preregistered* sweep of the consensus signal (3,520 rules, acceptance gates fixed in advance) clears **no gate**: staying accurate forces near-zero saving, and real saving forces a multi-point accuracy drop. The reason is directional—consensus forms *while the answer is still moving*, so a stop destroys a correct-in-the-end answer up to **35**$\times$ more often at an aggressive stop than it rescues a wrong one. The failure is specific to consensus: swept identically, a boundary-confidence method (DEER) **clears all three gates** and holds on a held-out split and two unseen models. Early exit is possible—just not on the consensus signal. We release the protocol, the frozen trajectories, and the full sweep.

1 Introduction

Reasoning language models (DeepSeek-AI, 2025; Qwen Team, 2025; OpenAI, 2024) can spend thousands of tokens on a single answer. *Early exit*—halting a chain of thought once its answer is settled—could cut that cost (Chen et al., 2024; Fu et al., 2024), but only if the stopping signal is safe. A widely used signal is *intermediate answer consensus*: at regular intervals a partial trajectory is probed for its current answer (e.g., by appending a short "`Final Answer:`" suffix) and generation is stopped once recent probes *agree* (Fu et al., 2024). The underlying assumption is simple: if the model repeatedly says x, it has probably finished reasoning about x. This paper asks whether that assumption is warranted: **is intermediate agreement a safe signal on which to stop?**

Our answer is **no**, and the reason is not a badly tuned threshold—it is the signal itself. Intermediate consensus[1] forms *while the answer is still moving*: on a frozen replay, stopping on consensus destroys a correct-in-the-end answer up to **35** times more often at an aggressive stop (Section 5). This is an *accuracy tax* that is a property of the reasoning trajectory, not of how one probes it, so no rule in the searched space escapes it. We make the claim hard to dismiss as "you did not tune it well" by searching the space exhaustively under commitment: on frozen trajectories from two models and three benchmarks we run a *preregistered* sweep of the consensus signal—a window size $\times$ share threshold family plus operational knobs, 3,520 rules—with problem-grouped splits and acceptance gates fixed in advance (Sections 3–4). The outcome does not depend on any single fragile estimate: **not one of the** $3{,}520$ **rules is both safe and saving—none clears any gate**, and even large windows only trade the drop away for zero saving. To show this is the *signal* and not early exit, we sweep one non-consensus method—DEER, which reads the model's confidence at a reasoning boundary—through the identical pipeline and gates: it clears all three, losing as little as 0.3 pp while saving 28%. The consensus frontier reproduces on a held-out test split (r=0.98) and on unseen models at 4$\times$ the scale and a different architecture (Section 4). Two published-signal checks bracket the contrast: a faithful Certaindex reproduction on the same trajectories collapses by 56–70 pp, while DEER clears the gates on the unseen models too. Figure 1 is the whole paper in one picture: the consensus rules that clear the gates in-sample on train leave the safe-and-saving corner on dev and test, while the three DEER operating points selected on dev stay in it out of sample.

[1] "False consensus" here denotes non-terminal intermediate probe agreement, unrelated to the social-psychology term (Ross et al., 1977); we study answer-level, not token/layer-level, early exit (Graves, 2016; Schuster et al., 2022).

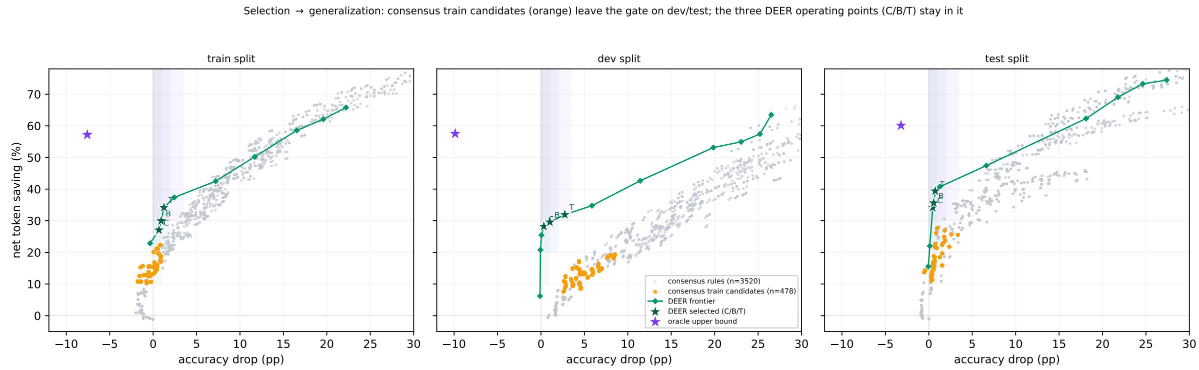

Figure 1: **Selection → generalization, in one picture** (two development models; macro-averaged over the environments; net token saving vs. accuracy drop). Grey: all 3,520 consensus rules. **Orange**: the 478 consensus rules that clear the conservative gate *in-sample on train*—on *dev* and *test* the same rules leave the shaded gate boxes (higher drop, lower saving), so the preregistered gate admits none out of sample. **Green**: the DEER threshold frontier (swept through the identical pipeline and accounting); the labelled stars are the three operating points selected on dev (**C**onservative/**B**alanced/**T**oken-efficient), which stay inside the gate on dev and test. **Purple star**: an oracle upper bound (earliest correct probe), far in the corner that neither method reaches. The failure is a property of the consensus *signal*, not of early exit.

Contribution. Our contribution is a rigorous, mechanism-backed demonstration that *intermediate answer-consensus is an unsafe early-exit signal for reasoning LLMs*, established so that it cannot be waved away as an untuned heuristic. Three components support this one claim:

- **A preregistered, probe-independent test** (frozen trajectories, offline re-probing, a two-parameter consensus family enumerated to 3,520 rules, and selection gates fixed in advance) under which *no* consensus rule is both safe and token-saving—and which we release in full (Sections 3–4).

- **A mechanism** that explains *why*: a directional "continued-reasoning-corrects" effect ($\gg$1:1) makes the accuracy cost of stopping intrinsic to the trajectory and independent of probe density, so the failure reflects the reasoning trajectory rather than the particular rules we searched (Section 5).

- **Out-of-distribution and cross-signal confirmation**: the consensus frontier reproduces on a held-out split and two unseen models, a published Certaindex reproduction collapses, and a non-consensus signal (DEER) swept under the identical protocol *clears the same gates* on the same trajectories—localizing the failure to consensus and indicating where a safe signal must come from (Sections 4–5).

2 Related Work

Probe-based / consensus early exit. The methods we scrutinize periodically query a partial reasoning trajectory for its current answer and halt on agreement. Dynasor / Certaindex (Fu et al., 2024) formalize this with a "certainty index" over recent probe answers and use it to terminate early when serving reasoning programs, one of several responses to the *overthinking* of reasoning models (Chen et al., 2024; Yang et al., 2025). We adopt the same probe mechanism (a short boxed-answer suffix) and notion of probe agreement, and ask how far *any* rule in this family can be pushed before it becomes unsafe.

Early-stopping self-consistency and self-correction. Probe-based early exit is the online, *single-trajectory* analogue of early-stopping self-consistency (Wang et al., 2023; Aggarwal et al., 2023; Li et al., 2024): it votes across snapshots of one in-progress trajectory rather than across finished samples. That difference is the crux—snapshots of an unfinished trajectory are not exchangeable draws from a stable answer distribution, because the trajectory is still correcting itself. This intrinsic self-correction differs from the prompted, post-hoc self-correction that Huang et al. (2024) find unreliable: we measure answer evolution *within* one uninterrupted trajectory, and find (Section 4) that a model's mid-trajectory answer is far from final.

Alternative termination signals. Rather than probe agreement, a stop can read other signals: verifier- and process-reward-guided decoding (Lightman et al., 2024; Wang et al., 2024) score partial reasoning, and adaptive-computation methods learn a halting signal at the token or layer level (Graves, 2016; Schuster et al., 2022). DEER (Yang et al., 2025) stops on the model's confidence in a trial answer at reasoning boundaries. We use DEER only as a *positive control* (Section 5): run through the identical pipeline and acceptance gates, it clears them where no consensus rule does, indicating that the failure is specific to the probe-agreement signal rather than to early exit itself. We do not claim this broader class of signals is categorically superior—our experiments isolate one signal, not a taxonomy of methods. Test-time scaling trades compute for accuracy via repeated sampling and search (Snell et al., 2024; Brown et al., 2024); early exit is its dual, spending less without losing accuracy.

Preregistration. To keep an exhaustive rule search from silently overfitting to a favorable operating point, we fix the rule schema, splits, and acceptance gates before looking at held-out data (Nosek et al., 2025). This preregistration (Section 3) is what makes a *negative* result interpretable.

3 Method

To ask whether *any* probe-based rule can succeed, we need (i) an evaluation that a rule cannot game by changing the reasoning it observes, and (ii) a selection procedure that cannot silently overfit to a lucky operating point. We preregister both.

3.1 Probe-independent, frozen trajectories

For each problem we generate exactly *one* main trajectory in a single request (caps: 16K tokens for MATH500 and AMC23, 32K for AIME24) and freeze it. All probing is then performed *offline* against fixed text prefixes of that frozen trajectory. Because probes never re-enter the decode, changing a probe schedule—or a stopping rule—cannot change the underlying reasoning; every rule is scored against the same trajectories. We build two offline probe banks: a *dense* bank that re-probes every 64 tokens with simple@32 (a 32-sample boxed-answer probe), and an *adaptive* bank that additionally fires event-triggered probes at entropy drops and at conclusion/reflection/answer markers.

3.2 Token accounting

For a rule that stops a problem at token position s after consuming probe output p, we charge total decode tokens $T = s + p$ and compare against the frozen baseline B (the full trajectory, capped at budget). *Net* savings is $(B - T)/B$ and *gross* savings is $(B - s)/B$; their gap is the probe cost. The accuracy drop compares the correctness of the committed answer at s against that of the frozen final answer (it is not an agreement rate between them). Two quantities therefore move independently, and we track both: *where* the rule stops (setting the accuracy drop and the main-token saving s) and *how much it probes* to get there (adding p). Section 5 shows that only the first is intrinsic.

3.3 The consensus signal: window size and share threshold

The consensus signal any probe-based rule can read reduces to two hyperparameters: a **window size** W (how many of the most recent probes to inspect) and a **share threshold** s (what fraction of that window must agree on a single answer before a stop is allowed). This pair spans the family to a close approximation: $W{=}1$ is the "latest-probe" rule, $s{=}1.0$ is strict unanimity over the last W probes (the Certaindex-style stop), and an entropy-over-the-window criterion closely tracks the same agreement and adds no materially new operating point on our grid. We sweep $W \in \{1, 3, 5, 8, 12, 16, 24, 30\}$ and $s \in \{0.6, 0.8, 1.0\}$, deliberately including *large* windows so a rule can demand a long, stable agreement before committing.

Around this core we vary only the operational knobs that govern *when* and *how* the signal is read: **probe schedule** (fixed interval $\in \{64, 128, 256, 512\}$ tokens, or event-triggered probing at conclusion/entropy/reflection markers); **maturity** (a minimum-token floor $\in \{0, 512, 1024, 2048, 4096\}$ before any stop is allowed); **validity** (an answer-shape filter, rejecting empty/letter answers or only empties); and **certainty** (optionally requiring the supporting probes to be flagged certain). Enumerating this preregistered grid over a fixed-schedule and an event-triggered family yields 3,520 candidate consensus rules; formal definitions are in Appendix A.

A better signal, swept identically. To test whether the failure is early exit *itself* or the consensus *signal*, we sweep one non-consensus method through the same pipeline, gates, and token accounting. **DEER** (Yang et al., 2025) stops on the model's own confidence in a trial answer at a reasoning boundary; its only hyperparameter is a confidence threshold, which we sweep over 14 values. We use the stronger *trial-answer-submit* variant (commit the trial answer directly, no separate readout) rather than the faithful readout reproduction—our aim here is not to reproduce prior work but to ask whether *any* rule can clear the gates. Placing DEER and consensus on one frontier under identical accounting is what makes the comparison a controlled test of the signal.

3.4 Models, benchmarks, splits

Development uses two models—DeepSeek-R1-Distill-Qwen-7B and Qwen3-8B—on three benchmarks (MATH500, AMC23, AIME24), each at seeds 42/43/44: 18 model×benchmark×seed environments. Problems are partitioned *by problem id* into 60/20/20 train/dev/test splits (MATH500 300/100/100, AMC23 24/8/8, AIME24 18/6/6), so no problem leaks across splits. Confirmation reserves seeds 45/46/47 plus two unseen models—DeepSeek-R1-Distill-Llama-8B and Distill-Qwen-32B—all evaluated on the *test* split only.

3.5 Preregistered operating points and gates

We fix three operating points and their acceptance gates *before* looking at held-out data (Table 1). Each gate is applied in order: first the **total accuracy drop** (macro-averaged over the 18 environments) must stay under a cap; then the **total net token saving** (macro-averaged) must clear a floor; and finally a **positive-saving fraction** (psf, the fraction of environments with strictly positive net saving) floor must be met. A rule is accepted at an operating point only if it satisfies all three on train+dev. The saving floor is what distinguishes a genuinely useful early exit from a rule that stays accurate only by almost never stopping.

3.6 Metric resolution

The total accuracy drop is a macro-average over the 18 environments (each model×benchmark×seed weighted equally). We span two models, three benchmarks, and three seeds and weight them equally so that selection rewards a rule that is robust across benchmarks, models, and randomness, not one that overfits a single setting. Pooling problems instead would let MATH500 (100 dev problems per

Operating point	Max total acc. drop	Min total saving	psf floor
conservative	1.0 pp	10%	0.80
balanced	2.0 pp	20%	0.80
token_efficient	3.5 pp	30%	0.70

Table 1: Preregistered acceptance gates. A rule must keep the macro-averaged accuracy drop under "max total acc. drop," deliver at least "min total saving" net tokens, and post positive net saving on at least a psf fraction of environments. Gates are fixed in advance and *not* relaxed post hoc.

seed vs. 8 for AMC23 and 6 for AIME24) dominate the headline, so a rule that wins only on MATH500 could pass; equal per-environment weight forces a selected rule to hold on the harder competition sets too. Macro also keeps any single problem from swinging the result—one AIME24 problem is 16.7 pp of a 6-problem split but only $\sim 1/108$ of the macro drop—while the three seeds damp the extra per-cell noise of the small sets. The gates are read on the dev split (the selection target) with train reported alongside; because *no* consensus rule clears any gate by a wide margin (Section 4), the finding does not rest on a single fragile cell. Held-out confirmation on the test split and two unseen models provides the complementary out-of-distribution check.

3.7 Preregistration commitments

Three commitments make the result interpretable. **(C1)** The test split and both confirmation models are never read during rule selection. **(C2)** All three operating points and their gates in Table 1 are fixed before seeing held-out data and are not relaxed afterwards; if no rule passes a gate, that is reported as the finding rather than engineered away, and all three points are reported regardless of outcome. **(C3)** The frozen rule set is fixed on train+dev and its hash recorded before any test/confirmation evaluation. Section 4 reports what happened when we applied C2.

4 Experiments

We first confirm the phenomenon the paper turns on—intermediate consensus is non-terminal (§4.1)—then report the preregistered sweep and its central negative result: *no* consensus rule is both safe and saving (§4.2). We then show the failure is specific to the *signal*, not to early exit, by sweeping a non-consensus method (DEER) through the identical pipeline and gates and finding that it *does* clear them (§4.3); a faithful Certaindex reproduction on the same trajectories collapses (§4.4). Finally we confirm the frontier out of distribution (§4.5).

4.1 False consensus is real

We log a dense probe stream for every MATH500 problem under DeepSeek-R1-Distill-Qwen-7B at the main 16K/32K budgets (all 500 problems, three seeds; 1,500 trajectories), never intervening—the probe is a separate forward pass that only observes the main trajectory (full setup in Appendix E)—and read off four facts.

(1) Final agreement is reliable; early agreement is not. By the time a trajectory finishes, agreement does track correctness: *cumulative* agreement across all probes is 97.8% correct (186 trajectories reach it), and even the final five-probe *window* is 90.4% correct (1,205). But an online rule cannot wait for the end—it fires the *first* time probes agree, mid-trajectory, where the answer is far from settled (facts 2–4). The operative question is not whether *final* consensus is trustworthy, but whether *early* consensus is—and it is not. **(2) Recovery is common.** Of the 1,148 trajectories whose *first* probe is wrong, 89.1% eventually flip to correct; of the 736 that formed a three-probe consensus *different* from their final answer, 84.2% ended up correct. **(3) Later consensus is worse.** Consensus formed before 512 tokens is 91.0% correct, but consensus forming only after 2048 tokens is 71.6% correct—hard problems converge slowly *and* to worse answers. **(4) The cost lands at the stop.** A naive stop (first three consecutive agreeing, certain, non-empty probes) fires on 1,477/1,500 trajectories and commits to answers correct only 50.5% of the time, whereas letting those same problems run to completion reaches 90.7%: a 40.2 pp loss.[2]

A good stopping rule must therefore be much more than "agreement." The rest of this section asks whether the *best possible* such rule, drawn from a large preregistered space, can be both safe and economical.

4.2 The safe-and-saving corner is empty for consensus

We replay all 3,520 consensus rules on the frozen trajectories and aggregate to per-environment metrics—126,720 rows $=$ 3,520 rules $\times$ 18 environments $\times$ 2 splits (train, dev)—then, per rule, its total (macro) accuracy drop and net token saving (accounting of Section 3.2). Applying the preregistered gates yields the paper's central negative result.

The outcome holds across environments: across all 126,720 rule–environment evaluations accuracy strictly *falls* in 64.9% and strictly *rises* in only 10.5% (unchanged in 24.6%), for a mean drop of 13.0 pp; over rules the total dev drop has median 13.2 pp and a minimum of 0.81 pp. **Not one of the** 3,520 **rules clears any gate.** The reason is the shape of the frontier (dev panel of Figure 1): staying accurate and saving tokens are in direct tension. Only 40 rules keep the total drop at or below the conservative 1.0 pp cap, and the most any of them saves is 0.2%—far under the 10% floor; conversely, the first rule to save 10% already costs 2.66 pp, saving 20% costs 6.17 pp, and saving 30% costs 11.8 pp. Crucially, the sweep now includes *large* windows (up to $W{=}30$): these do buy low accuracy drop, but only by stopping so late that net saving collapses to zero. No setting of the consensus signal escapes the trade-off; the safe-and-saving corner is empty.

4.3 Early exit *is* possible—with a different signal

The same gates are cleared, comfortably, by a rule that does *not* read probe agreement. Sweeping DEER's confidence threshold through the identical pipeline and token accounting (Section 3.3), its trial-answer-submit variant passes all three operating points on dev: the conservative point loses only 0.33 pp while saving 28.2%, the balanced point 1.03 pp at 29.6%, and the token-efficient point 2.75 pp at 31.9%; at a stricter threshold it is accuracy-neutral (-0.06 pp) while still saving 20.8%. In the dev panel of Figure 1 the DEER frontier sits *above* the consensus frontier everywhere in the safe region and enters every gate box that no consensus rule reaches. Early exit is therefore not impossible; the consensus *signal* is what cannot make it safe (Table 2).

4.4 Consensus in the wild: a faithful Certaindex reproduction collapses

The sweep shows no *tuned* consensus rule is safe; a published consensus method shows the same failure at its shipped operating point. We reproduce three prior stoppers in a common harness—

[2] A simplified heuristic, *not* the Certaindex algorithm (§4.4); it fires early because the first stable, certain agreement is typically a mid-trajectory placeholder. The swept rules below and the mechanism of Section 5 quantify the same directional accuracy tax across the full rule space.

Method	Acc.	Δacc (macro)	Net saving
Full generation	82.5%	—	0%
Consensus (drop$\leq$ 1.0)	81.6%	-0.9 pp	+0.2%
Consensus (save$\geq$ 10%)	79.9%	-2.7 pp	+10.7%
DEER (conservative)	82.5%	-0.33 pp	+28.2%
DEER (balanced)	81.9%	-1.03 pp	+29.6%
DEER (token_eff.)	80.1%	-2.75 pp	+31.9%

Table 2: Dev operating points (macro over 18 environments). No consensus rule is simultaneously safe ($\leq$ 1.0 pp) and saving ($\geq$ 10%): capping the drop forces saving to near zero, and demanding saving forces a large drop. DEER, reading a non-consensus signal, clears all three gates on the same trajectories under the same accounting.

the same frozen trajectories, the same simple@32 probe bank, the same token accounting. **CertaIndex** (Fu et al., 2024) is the faithful share-threshold stop (the direct descendant of the consensus family); **DEER** (Yang et al., 2025) here is the *faithful* readout reproduction (the stronger trial-answer-submit variant is what we swept in §4.3); **TJE** (Anonymous, 2026) is a token-level stop. Table 3 reports the reproductions (not the original papers' end-to-end runs).

CertaIndex, at 99.8% stop rate and 70 pp lost, is the accuracy tax paid in full—the shipped consensus method landing exactly where the swept consensus frontier says it must. TJE is unsafe on the weaker model. Even the *faithful* DEER readout, a non-consensus signal, already avoids collapse; the stronger trial-answer-submit variant we swept turns that into a gate-clearing frontier (§4.3). That only the non-consensus signal survives is the crux we analyze in Section 5.

4.5 The frontier reproduces out of distribution

We read the test split once (commitment C3). Because the development result is that *no* consensus rule clears any gate, there is nothing to confirm by re-selection; we ask whether the *frontier* reproduces, ranking rules on dev and measuring on test.

Held-out test split. Each consensus rule's total drop on test tracks its dev value closely ($r = 0.98$). The split-invariant claim is the empty *joint* gate: **no consensus rule clears the conservative gate on both splits**—0 on dev, and the 444 rules admissible on *test alone* are exactly the in-sample winners the held-out split exists to reject (all fail on dev). DEER, by contrast, remains admissible across splits: 3 of its thresholds clear the conservative gate on *both* dev and test (and 5 clear balanced on both), so the cross-signal contrast is not a dev artifact.

Unseen scale (4×) and architecture. Both unseen models are evaluated on the test split at three seeds (45/46/47; nine environments each), so their gates are as well resolved as development's. On Distill-Qwen-32B, never used in development, the consensus frontier reproduces ($r = 0.97$ against dev drop); on the held-out architecture and family DeepSeek-R1-Distill-Llama-8B it reproduces at $r = 0.94$.[3] The *conservative* gate stays empty on every model—dev, 32B, and Llama all admit 0 rules (the best rule under a 1.0 pp drop saves 0.6% on 32B and 9.3% on Llama). At the looser gates a scale effect appears: on the 4× larger 32B a few consensus rules become admissible in-sample (4 at balanced, 6 at token-efficient), reflecting that a bigger model's intermediate answers stabilize somewhat earlier—but these are not the dev-selected rules (dev admits none), and on the same-scale Llama architecture the gates stay empty (0/0/0). Consensus thus yields no rule that is safe-and-saving where it is selected, on any model. DEER, swept identically on these unseen models, **clears the gates on both**: on Qwen-32B it is accuracy-neutral at the conservative point (-0.24 pp while saving 32.4%, τ=0.97) and on Llama-8B it loses 0.67 pp at 26.7% (τ=0.99), passing all three gates. Consensus is unsafe, and DEER safe-and-saving, across all four axes we vary—benchmark, seed, 4× scale, and architecture/family.

Figures 2 and 3 show this out of sample, per model and per benchmark, by tracking the *same* strategies onto each test panel: the consensus rules selected as gate-clearing on train (orange) and the three DEER operating points selected on dev (labelled stars). Across scale (Qwen-32B), architecture (Llama-8B), and every benchmark, the orange consensus candidates scatter to low saving or higher drop—they do not transfer—while the dev-selected DEER points land back inside the gate on the test data. Each panel also plots an *oracle* upper bound—a perfect probe-based stop that commits the earliest correct probe—which sits far in the safe-and-saving corner (a negative drop of 2–

[3] The Llama-8B trajectories were re-collected after correcting a chat-template defect (a missing model-specific beginning-of-sequence token that degraded its generations); the corrected model scores ∼88% on MATH500.

Method	Model	Accuracy	Δacc	Fair token saving	Stop rate	Signal
Full generation	Qwen3-8B	85.4%	—	0%	0%	—
Full generation	DeepSeek-R1-Distill-Qwen-7B	79.8%	—	0%	0%	—
CertaIndex	Qwen3-8B	15.3%	−70.1 pp	90.1%	99.8%	consensus
CertaIndex	DeepSeek-R1-Distill-Qwen-7B	23.9%	−55.9 pp	76.7%	98.7%	consensus
TJE	Qwen3-8B	85.0%	−0.4 pp	2.0%	22.5%	token-level
TJE	DeepSeek-R1-Distill-Qwen-7B	60.7%	−19.1 pp	65.0%	93.4%	token-level
DEER	Qwen3-8B	86.2%	**+0.8** pp	16.3%	41.4%	boundary conf.
DEER	DeepSeek-R1-Distill-Qwen-7B	74.9%	−4.8 pp	20.2%	56.1%	boundary conf.

Table 3: Prior early-stoppers reproduced on our frozen trajectories (dev, macro over three benchmarks; probe tokens charged). CertaIndex—the consensus descendant—**collapses** (−56 to −70 pp) while saving 77–90% of tokens: it fires on almost every problem, very early, exactly the false-consensus trap of §4.1. **DEER**, which reads a *different* signal (boundary confidence), is the only method that stays near full accuracy—+0.8 pp on Qwen3-8B—while still saving tokens. We interpret this contrast in Section 5.

5 pp at 40–80% saving) that neither consensus nor DEER reaches: headroom that a better *realizable* signal might still recover. (On the small AIME24 set the frontiers are noisier and intertwine.)

5 Analysis

Section 4 is an existence proof over 3,520 rules, two seen and two unseen models, and both splits. This section explains *why* it holds and why it generalizes beyond the exact rules we searched: the accuracy cost of stopping on consensus is a directional property of the reasoning trajectory itself, not of a threshold or of probe density.

5.1 Stopping destroys more than it saves

Consider only the stops where committing early *changes* correctness relative to the trajectory's own final answer, and count two outcomes: (final-correct, stop-wrong)—a correct recovery destroyed—versus (final-wrong, stop-correct)—a wrong answer luckily banked. Pure sampling noise predicts a $\approx$ 1:1 ratio; a genuine "continued reasoning corrects" effect predicts $\gg 1$. On the frozen replay this ratio depends strongly on the window: it falls from $\approx$45:1 for a latest-probe stop ($W{=}1$) to $\approx$ 2:1 at the largest window ($W{=}30$). But the low end is bought only by stopping so rarely that net saving vanishes ($W{=}30$ fires on 121 dev problems vs. 668 at $W{=}1$); at any window that actually saves tokens the ratio stays well above 1:1. DEER, by contrast, holds $\approx$2.4–3.5:1 at its three operating points *while* saving 28–32%, because its boundary-confidence signal commits mainly once the trajectory has settled. Stopping on consensus therefore removes a correct-in-the-end answer many times more often than it rescues a wrong one—directional, not sampling noise. This is the recovery phenomenon of §4.1 (84.2%, 89.1%), now at the main 16K/32K budgets and as a signed effect.

5.2 The accuracy tax is intrinsic to the trajectory

Why is the ratio structural? For a rule stopping at token position s, the accuracy drop depends only on s: committing at s forgoes whatever correction the trajectory would have made after it. It is a property of the reasoning, not of the probe, so it survives *any* probing scheme. This is what never reaches zero in the sweep. The probe grid (intervals 64/128/256/512 tokens) makes essentially every stop position reachable and the observed floor is the minimum over these schedules; a sparser schedule can only stop later or at a subset of positions, never at a safer one the grid missed. Hence no threshold on the consensus signal *within the searched space* escapes the accuracy tax; the searched space is a lower bound on the cost, not a lucky miss. This does not turn a finite rule search into a universal impossibility theorem—it motivates caution for consensus variants beyond the preregistered schema rather than proving a universal claim about them. In short, agreement reports what the model has said recently, not whether its reasoning has finished.

5.3 Ruling out the "it is just dense probing" objection

The natural rebuttal is that the negative result is an artifact of charging dense re-probing. It is not, and separating two costs shows why. Besides the accuracy tax, a rule pays a *probe tax*: the probe output p (Section 3.2). Under dense re-probing a late-stopping large-window rule pays p over a long prefix, opening a gap between its *gross* savings and

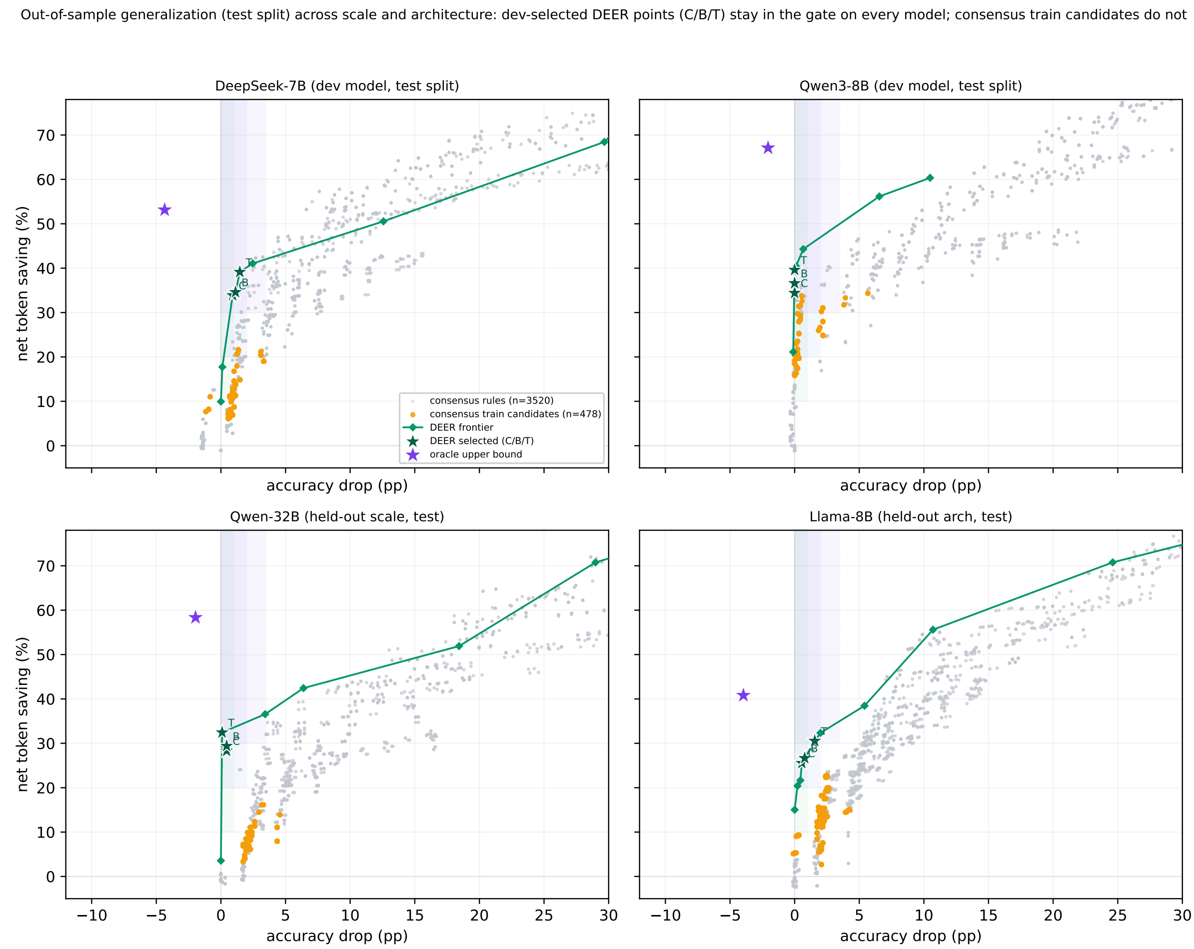

Figure 2: **Out-of-sample generalization across scale and architecture** (all panels test split). Grey: consensus rules; **orange**: consensus train candidates; **green**: DEER frontier with the three dev-selected operating points (C/B/T); **purple**: oracle upper bound. The two development models (top) are shown on their held-out test seeds; the two held-out models (bottom) are entirely unseen. The dev-selected DEER points stay inside the conservative gate on every model; no consensus rule does.

Consensus point	Gross saving $(B-s)/B$	Net saving $(B-s-p)/B$
save$\geq$ 10% (drop 2.7 pp)	+14.7%	+10.7%
save$\geq$ 20% (drop 6.2 pp)	+27.0%	+20.2%

Table 4: Gross vs. net savings for two consensus frontier points; the gap ($\sim$ 4–7 pp) is the probe tax. It shifts the *savings* axis but never rescues the accuracy drop—which is the intrinsic tax that keeps the corner empty.

its (smaller) *net* savings (Table 4). Unlike the accuracy tax, the probe tax scales with probe density and is reducible (sparser or shorter probes, KV-cache reuse). The two are separable: the *savings* axis is partly a probe-tax artifact and is addressable, but the *accuracy* floor is probe-independent and is what makes the corner empty—and it is the accuracy floor, not the probe tax, that DEER escapes.

5.4 It is the signal, not early exit

The joint sweep of §4.3 localizes the cause. Every method that stops on *probe agreement*—the swept consensus family and CertaIndex alike—either fails a gate or collapses outright; CertaIndex's 70 pp loss at 99.8% stop rate is the accuracy tax paid in full. The method that clears the gates, DEER, does not read probe agreement at all: it stops on the model's own confidence in a trial answer at a reasoning boundary, and in Figure 1 its threshold frontier sits *above* the consensus Pareto boundary throughout the safe region. Because DEER is swept through the identical pipeline, gates, and token accounting, the contrast is a controlled test: the same early-exit machinery succeeds on a forward-looking signal and fails on the consensus signal. The failure is a property of the *signal*, not of early exit.

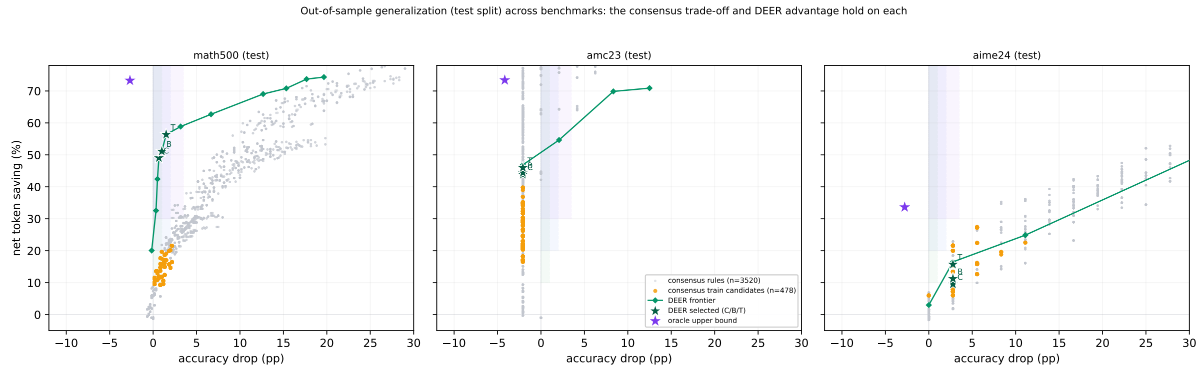

Figure 3: **Out-of-sample generalization across benchmarks** (test split, development models). The consensus trade-off and the DEER advantage hold on MATH500 and AMC23; on the small AIME24 set (few problems per environment) the two frontiers are noisier and intertwine. Markers as in Figure 2.

5.5 Implication: change the signal, not the threshold

Much prior effort tunes consensus-based stops (Fu et al., 2024; Aggarwal et al., 2023; Li et al., 2024); our frontier shows the ceiling of that approach is low, and the mechanism above says no amount of tuning raises it. The constructive implication is that a safe signal should be about whether the *trajectory itself* will still change, not about the agreement of backward-looking probes. Our joint sweep already demonstrates this: the boundary-confidence signal (DEER) clears the same gates the whole consensus family fails (§4.3). Verifier- and process-reward-based termination (Lightman et al., 2024; Wang et al., 2024) and learned halting (Graves, 2016; Schuster et al., 2022) point the same way. Turning any of these into a *confirmed* method—with its own train/dev→test validation—is the natural next paper; our contribution is to establish rigorously which signal not to build it on. Two caveats bound the claim: it concerns *consensus-based* rules within the searched schema and dense-probe regime, and it does not say early exit is impossible—only that this signal cannot make it safe.

6 Conclusion

We asked whether intermediate answer-consensus is a safe signal on which to early-exit a reasoning LLM. It is not, and not because the rule is untuned: intermediate consensus forms while the answer is still moving, so stopping on it destroys a correct-in-the-end answer up to about 35 times more often (at an aggressive stop) than it rescues a wrong one. This accuracy tax is a property of the trajectory rather than the probe, and it shows up as an empty safe-and-saving corner across a preregistered space of 3,520 rules—not one of which clears any gate—reproduced on a held-out split and two unseen models, and paid in full by a faithful Certaindex reproduction that collapses by 56–70 pp. That the corner is empty for consensus but not for early exit is the point: swept through the identical protocol, a method reading the model's own boundary confidence (DEER) clears all three gates on the very same trajectories, so the failure is specific to consensus. The takeaway for anyone building safe early exit is therefore about the choice of signal: read the model's own forward-looking state, not the agreement of its backward-looking probes.

Limitations

Scope of the negative result. The central result—no consensus rule clears any gate—is established on 18 development environments (two models, three benchmarks, seeds 42/43/44) and confirmed on the held-out test split and two unseen models (Section 4). It is a statement about the space searched: one boxed-answer probe suffix (simple@32) and one preregistered consensus schema (3,520 rules over a window-size $\times$ share-threshold family plus operational knobs). Other probe wordings and signals outside this schema are not searched; the mechanism (Section 5) is our argument that consensus variants would not help, but it is an argument, not an exhaustive proof.

DEER as the positive control, not a benchmarked SOTA. We sweep DEER's trial-answer-submit variant to establish that *some* signal clears the gates the consensus family fails (on the development models, the held-out test split, and both unseen models); we do not claim it is the best possible early-exit method. Its role is a controlled positive control for the signal, not a leaderboard entry.

Small held-out sets on hard benchmarks. AIME24 and AMC23 are small (dev splits of 6 and 8 problems per seed), and the unseen models contribute a single seed, so those per-environment cells are noisy. The total drop macro-averages over the 18 development environments (three seeds per benchmark$\times$model), and we read the single-seed unseen models as frontier-reproduction evidence rather than as independent gate tests (Section 3.6).

Probe-tax dependence of the savings axis. Net-savings numbers charge dense re-probing (every 64 tokens, 32-token probes); a sparser or KV-cache-reusing schedule would change the *savings* axis and could move some rules to positive net savings. We separate gross from net savings (Section 5) precisely so the probe-tax-dependent part is distinguishable; the accuracy-tax result does not depend on the schedule.

Domain. All benchmarks are competition mathematics with checkable final answers. Whether false consensus and the accuracy tax transfer to open-ended reasoning, code, or agentic tasks is out of scope.

References

Pranjal Aggarwal, Aman Madaan, Yiming Yang, and Mausam. 2023. Let's sample step by step: Adaptive-consistency for efficient reasoning and coding with LLMs. In *Proceedings of EMNLP*.

Anonymous. 2026. Token-level justification for early-exit in reasoning. In *Findings of EACL*.

Bradley Brown, Jordan Juravsky, Ryan Ehrlich, Ronald Clark, Quoc V. Le, Christopher Ré, and Azalia Mirhoseini. 2024. Large language monkeys: Scaling inference compute with repeated sampling. *arXiv preprint arXiv:2407.21787*.

Xingyu Chen, Jiahao Xu, Tian Liang, Zhiwei He, Jianhui Pang, Dian Yu, Linfeng Song, Qiuzhi Liu, Mengfei Zhou, Zhuosheng Zhang, Rui Wang, Zhaopeng Tu, Haitao Mi, and Dong Yu. 2024. Do not think that much for 2+3=? on the overthinking of o1-like LLMs. *arXiv preprint arXiv:2412.21187*.

DeepSeek-AI. 2025. DeepSeek-R1: Incentivizing reasoning capability in LLMs via reinforcement learning. *arXiv preprint arXiv:2501.12948*.

Yichao Fu, Junda Chen, Siqi Zhu, Zheyu Fu, Zhongdongming Dai, Aurick Qiao, and Hao Zhang. 2024. Efficiently serving LLM reasoning programs with Certaindex. *arXiv preprint arXiv:2412.20993*.

Alex Graves. 2016. Adaptive computation time for recurrent neural networks. *arXiv preprint arXiv:1603.08983*.

Dan Hendrycks, Collin Burns, Saurav Kadavath, Akul Arora, Steven Basart, Eric Tang, Dawn Song, and Jacob Steinhardt. 2021. Measuring mathematical problem solving with the MATH dataset. In *NeurIPS Datasets and Benchmarks Track*.

Jie Huang, Xinyun Chen, Swaroop Mishra, Huaixiu Steven Zheng, Adams Wei Yu, Xinying Song, and Denny Zhou. 2024. Large language models cannot self-correct reasoning yet. In *International Conference on Learning Representations (ICLR)*.

Yiwei Li, Peiwen Yuan, Shaoxiong Feng, Boyuan Pan, Xinglin Wang, Bin Sun, Heda Wang, and Kan Li. 2024. Escape sky-high cost: Early-stopping self-consistency for multi-step reasoning. In *International Conference on Learning Representations (ICLR)*.

Hunter Lightman, Vineet Kosaraju, Yura Burda, Harri Edwards, Bowen Baker, Teddy Lee, Jan Leike, John Schulman, Ilya Sutskever, and Karl Cobbe. 2024. Let's verify step by step. In *International Conference on Learning Representations (ICLR)*.

Brian A. Nosek et al. 2025. Preregistration in machine learning research. *arXiv preprint*.

OpenAI. 2024. Learning to reason with LLMs. `https://openai.com/index/learning-to-reason-with-llms/`.

Qwen Team. 2025. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*.

Lee Ross, David Greene, and Pamela House. 1977. The "false consensus effect": An egocentric bias in social perception and attribution processes. *Journal of Experimental Social Psychology*, 13(3):279–301.

Tal Schuster, Adam Fisch, Jai Gupta, Mostafa Dehghani, Dara Bahri, Vinh Q. Tran, Yi Tay, and Donald Metzler. 2022. Confident adaptive language modeling. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Charlie Snell, Jaehoon Lee, Kelvin Xu, and Aviral Kumar. 2024. Scaling LLM test-time compute optimally can be more effective than scaling model parameters. *arXiv preprint arXiv:2408.03314*.

Peiyi Wang, Lei Li, Zhihong Shao, Runxin Xu, Damai Dai, Yifei Li, Deli Chen, Yu Wu, and Zhifang Sui. 2024. Math-shepherd: Verify and reinforce LLMs step-by-step without human annotations. In *Proceedings of ACL*.

Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc Le, Ed Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. 2023. Self-consistency improves chain of thought reasoning in language models. In *International Conference on Learning Representations (ICLR)*.

Chenxu Yang, Qingyi Si, Yongjie Duan, Zheliang Zhu, Chenyu Zhu, Zheng Lin, Li Cao, and Weiping Wang. 2025. Dynamic early exit in reasoning models. *arXiv preprint arXiv:2504.15895*.

A Rule Schema Details

The consensus schema of Section 3 centers on two hyperparameters and a few operational knobs, enumerating to 3,520 candidate rules over a fixed-schedule and an event-triggered family. Each rule fixes:

1. **Window size** $W \in \{1, 3, 5, 8, 12, 16, 24, 30\}$: the number of most recent probes inspected. $W{=}1$ is the latest-probe rule.

2. **Share threshold** $s \in \{0.6, 0.8, 1.0\}$: the fraction of the last W probes that must agree on one answer (share measured over the full window, so an empty or dissenting probe counts against consensus); $s{=}1.0$ is strict unanimity. An entropy-over-the-window criterion closely tracks the same agreement and adds no materially new operating point on our grid, so it is not swept separately.

3. **Probe schedule**: fixed interval $\in \{64, 128, 256, 512\}$ tokens, or `adaptive_event` with entropy-drop and marker triggers and a fallback interval.

4. **Maturity**: a minimum-token floor $\in \{0, 512, 1024, 2048, 4096\}$ before any stop is allowed.

5. **Certainty**: optionally require the supporting probes flagged certain (minimum fraction).

6. **Validity**: answer-shape filter (reject empty / single-letter answers to suppress the Type-E probe artifact of Appendix E), or accept any non-empty answer.

Certainty. A probe is *certain* when its boxed answer is emitted with high token-level confidence: concretely, the mean probability of the answer tokens under the probe forward pass exceeds a fixed threshold (the Certaindex certainty flag). For the simple@32 bank this is aggregated over the 32 samples as the fraction agreeing on the modal answer. Dimension 5 gates a stop on the fraction of the supporting probes that are certain.

Entropy triggers. The `adaptive_event` schedule fires an extra probe when the token-level predictive entropy of the main decode drops by at least ΔH relative to a trailing baseline (an "the model just committed" signal), and at conclusion/reflection/answer marker tokens (e.g., "therefore", "the answer is"), subject to a minimum inter-probe gap and a fallback interval.

B Preregistration Text

The protocol version, split manifest hash, candidate-rule generation script, and the three operating-point gates (Table 1) were fixed before any held-out evaluation. Commitments C1–C3 (Section 3) were recorded with the protocol. All three operating-point gates were fixed in advance; when the conservative gate proved empty we followed C2 and report the empty gate as the finding rather than relaxing it, and all three points are reported for both consensus and DEER regardless of outcome.

C Reproducibility

Frozen trajectories, offline probe banks, the sweep archive (126,720 dev metric rows plus the DEER threshold sweep), and the aggregation scripts are released. Tables 5–6 were reproduced to the reported precision directly from the released sweep archive. Grading uses a robust answer-equivalence check that tries the raw and normalized reference forms and both string and math-expression equality; per-problem correctness for the frozen baseline is recomputed at replay time rather than trusted from collection.

D Window-Size and DEER Frontiers

Table 5 traces the consensus trade-off directly along the window axis: as the window W grows, the minimum attainable accuracy drop falls monotonically (from 22 pp at $W{=}1$ to 0.8 pp at $W{=}30$), but the maximum net saving falls with it (from 98% to 39%)—the low-drop region is reachable only at large windows, where saving has already collapsed. This is why no single (W, s) setting lands in the safe-and-saving corner. Table 6 gives the DEER threshold frontier that does reach it.

E False Consensus: Full Analysis

Section 4.1 summarizes, in four facts, what this appendix establishes in full: in an intervention-

Window W	Min total drop (pp)	Max net saving
1	22.25	98.1%
3	11.76	93.8%
5	8.01	87.9%
8	4.91	72.9%
12	1.85	62.0%
16	1.74	55.1%
24	1.69	43.2%
30	0.81	38.8%

Table 5: Consensus frontier along the window axis on dev (over all share thresholds and operational knobs at each W). Larger windows buy a smaller minimum accuracy drop but only by stopping later, so the maximum net saving falls in lockstep; the two never meet in the gate region.

Threshold τ	Total drop (pp)	Net saving
0.9999	−0.06	20.8%
0.999	0.06	25.4%
0.995	0.33	28.2%
0.99	1.03	29.6%
0.97	2.75	31.9%
0.95	5.87	34.8%
0.90	11.43	42.6%

Table 6: DEER trial-answer-submit threshold frontier on dev (macro over 18 environments, same token accounting). Unlike consensus, low drop coincides with substantial saving—the conservative/balanced/token-efficient gates are all cleared around $\tau \in [0.97, 0.995]$.

free setting, intermediate probe consensus is not a reliable stopping signal.

E.1 Setup

We analyze the frozen main trajectories (Section 3): DeepSeek-R1-Distill-Qwen-7B on all 500 MATH500 problems (Lightman et al., 2024; Hendrycks et al., 2021) at the main 16K-token budget, temperature 0.6, top-p 0.95, across three seeds (1,500 trajectories in total).[4] The dense probe bank appends the boxed-answer probe suffix every 64 generated tokens and reads off the model's current answer. The probe is a separate forward pass: the controller only *observes*, never halting or altering the main trajectory. This yields, per trajectory, the full answer sequence a stopping rule would have access to at the budget used throughout the paper.

[4]Train/dev problem ids are collected at seeds 42/43/44 and the held-out test ids at seeds 45/46/47; their union covers all 500 problems. This is a descriptive characterization of the frozen trajectories—it selects and tunes nothing—so it does not touch the preregistered sweep or the test-split commitment.

E.2 Final agreement is reliable; early agreement is not

Agreement measured at the *end* of a trajectory does track correctness. On the 186 trajectories that reach unanimous *cumulative* agreement (all non-empty probes concur), 97.8% are correct—only four whole-trajectory false consensuses; and of the 1,205 whose *final five-probe window* is unanimous, 90.4% are correct. But this end-state agreement is not what an online stopping rule can act on: a rule fires the *first* time it sees agreement, mid-trajectory, long before the answer has settled. The rest of this appendix shows that *early* agreement—the signal a controller must actually use—is far from terminal.

E.3 The cost lands at the stopping decision

We simulate a *naive consecutive-agreement* stop: halt at the first point where three consecutive probes agree, are non-empty, and are *certain* (the model emits the boxed answer with high token-level confidence; formal definition in Appendix A). This is a simplified heuristic, *not* the Certaindex algorithm, which we reproduce faithfully in §4.4. It fires on 1,477/1,500 trajectories. The committed answers are correct only 50.5% of the time, whereas letting those same problems run to completion reaches 90.7%—a 40.2 **pp accuracy loss** in exchange for an average saving of 3301 tokens, with 731 trajectories stopped on an outright wrong answer.

E.4 Recovery is common, and early stopping kills it

The loss is driven by *recovery*: continued reasoning overturns the intermediate answer far more often than intuition suggests. Of the 736 trajectories that at some point formed a three-probe consensus *different* from their final answer, 620 (84.2%) ended up correct; and of the 1,148 whose *first* probe was wrong, 89.1% eventually flipped to correct. Neither "first impression" nor "local consensus" is close to terminal.

Counterintuitively, *later* consensus is *less* trustworthy: consensus formed before 512 tokens is 91.0% correct, but consensus that only forms after 2048 tokens is 71.6% correct. Hard problems both converge slowly and converge to worse answers—exactly the problems on which an aggressive stopping rule does most damage.

E.5 An error taxonomy

Hand-inspecting stopped-but-wrong cases (a preliminary AI-assisted labeling of 28 cases from an earlier exploratory pass, not the automated counts above) yields *four* non-empty types out of five categories. By count: **(A)** numeric collapse—stable convergence to a wrong number—$14/28$; **(D)** derivation gaps—dropped roots/cases, unverified solutions—$7/28$; **(E)** format/option hallucination—a non-multiple-choice problem stably emitting a letter such as “B”—$6/28$, which is an artifact of the probe mechanism itself rather than a model belief; and **(B)** expression collapse—$1/28$. Type **(C)** (sign errors) did not occur in this sample. Type E is itself a methodological finding: roughly one in five “false consensuses” is a probe formatting artifact, implying that any deployed rule should filter answer *shape*.

Takeaway. These observations—local consensus unreliable, recovery common, late consensus worse—suggest that a good stopping rule must be much more than “agreement.” The rest of the paper asks whether the *best possible* such rule, drawn from a large preregistered space, can be both safe and economical. In the searched space, it cannot.